USN | | | | | | | | | | [22CSE316]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

PYTHON FOR DATA SCIENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5x2=10**
1.	**a.**	Define Fancy Indexing.	L1	CO1	**02**
	b.	Mention the usage of handling missing data.	L2	CO2	**02**
	c.	Write python program to plot scattered for the labels men and women.	L4	CO3	**02**
	d.	Compare loc[] and iloc[] in Pandas.	L3	CO2	**02**
	e.	Define Exploring data Analysis?	L1	CO2	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain in detail about numpy array slicing with an example.	L4	CO1	**05**
		OR			
	b.	Explain in detail about aggregate functions in Numpy with an example.	L4	CO1	**05**
3.	**a.**	Discuss the process of handling missing values in pandas with an example code.	L2	CO2	**05**
		OR			
	b.	Write a Python code to use append() and concat() functions for the high performance Pandas operations.	L4	CO2	**05**
4.	**a.**	Explain the purpose of customizing plot legends in Matplotlib with an example.	L3	CO3	**05**
		OR			
	b.	Differentiate plt.contour() and plt.contourf()? Explain in detail.	L2	CO3	**05**
5.	**a.**	Explain the purpose of the HashingVectorizer in Scikit-learn.	L2	CO3	**05**
		OR			
	b.	How does multicore parallelism improve performance in Scikit-learn	L2	CO3	**05**
6.	**a.**	How does NumPy Joining Array works. Explain with an example.	L4	CO4	**05**
		OR			
	b.	Explain in detail about universal functions in Numpy.	L4	CO4	**05**
7.	**a.**	Explain the difference between variance and standard deviation in descriptive statistics.	L2	CO5	**05**
		OR			
	b.	Describe the concept of covariance and its relationship with correlation.	L4	CO5	**05**

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		PART - C			6x10=60
8.	**a.**	Explain the concept of NumPy searching Arrays with an example code.	L2	CO1	**10**
		OR			
	b.	How does NumPy's broadcasting work? Explain with an example.	L2	CO3	**10**
9.	**a.**	Explain in detail about Combining Datasets- Merge and Join. Write Python code to perform merge and join on a pandas data set.	L4	CO3	**10**
		OR			
	b.	Write Python snippet to handle missing values in a dataset.	L4	CO4	**10**
10.	**a.**	Write Python snippet To remove duplicate values in a dataset.	L4	CO4	**10**
		OR			
	b.	Explain Data Indexing and selection in detail.	L2	CO2	**10**
11.	a.	Illustrate Three-Dimensional Plotting in Matplotlib, Geographic Data with Basemap, in detail.	L3	CO2	**10**
		OR			
	b.	Explain in detail about Pie chart. Write Python code to create a pie chart of the popularity of programming Languages.	L4	CO3	**10**
12.	**a.**	Describe Considering Timing and Performance in detail.	L2	CO2	**10**
		OR			
	b.	Define Multiprocessing. Write Python Code for Multiprocessing	L4	CO2	**10**
13.	**a.**	Write a Python code to perform descriptive statistics, Handling non categorical data, Boxplot, Covariance and correlation.	L4	CO5	**10**
		OR			
	b.	Describe Inspecting boxplots, Performing t-tests after boxplots in detail.	L2	CO4	**10**